

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Impacts of agriculture, social/political factors & sustainable agriculture - Set 1

Mark scheme

AQA spec 3.5.1.3-3.5.1.5; Biological resources; textbook pp. 328-337

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive • Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light; diversity falls • Hedgerow removal destroys habitat / corridors and nesting sites; loss of natural pest predator habitats increases pest damage; larger fields increase wind erosion; genetic contamination and introduced species also accepted	6
01.2	• Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows • These communities depend on the farming continuing	2
01.3	• CO₂ from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery; ploughing increases soil respiration • Methane from livestock intestines and rice paddies; nitrous oxide from manure and nitrogen fertilisers	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	
02.1	• Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala); increases evaporation • Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase • Evapotranspiration changes: increased in irrigated arid areas, decreased where forests were cleared	3

Question	Marking guidance	Marks
02.2	- Seasonal / local: avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes - Organic: uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species; free-range and Fairtrade also accepted for welfare / social reasons	4
02.3	- Food shortages after 1945 (dependence on US food aid); farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock - Guaranteed market / price support: government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output - Consequences: loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution; later food surpluses (butter mountains) that could not be sold without harming LEDC producers	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
03.1	- Other MEDCs (USA, Canada, Australia) also had surpluses; Japan already had suppliers - Eastern Europe / USSR needed food but could not pay - Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production	3
03.2	- Quotas limiting production (eg milk) - Set-aside payments for taking land out of production - Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) - Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming	4
03.3	<i>Any 3 from:</i> - Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape - Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat - Beetle banks of tussocky grass - natural pest predators - Skylark nesting plots; wild bird seed / nectar plants - winter food and pollinators - In-field tree protection; controlled stocking density to reduce compaction; ditch management; dry-stone walls; woodland management	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	

Question	Marking guidance	Marks
04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports • Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling • Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control • Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes • Social: control of impacts on rural communities, Fairtrade, local food; consumer choices • Evaluation: yields vs sustainability, cost, need for policy support; judgement	9
04.2	• Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
05.1	<i>Any 2 from:</i> • Hedgerows • Hay meadows • Grazed moorland / heathland • Chalk grassland	2
05.2	• Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers	2
05.3	• The introduced (non-indigenous) predator may eat other species as well as the pest	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
06.1	• Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed • Yield mapping of different parts of a field controls fertiliser application in each area for future crops • This maximises yield while avoiding wastage / reducing leaching and nutrient pollution	3
06.2	• Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive • They are mainly available in affluent societies; poorer farmers cannot afford the inputs	2
06.3	• Provides an income that allows producers to afford basic rights such as water, education, health care and food	1
06.4	• Nitrates are leached into water bodies used as sources of potable water • High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic	2
06.5	<i>Any 2 from:</i> • Anaerobic digestion by microbes in rice padi fields • Livestock intestines (ruminant digestion) • Anaerobic decay of manure / slurry	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	
07.1	• About 10% of land is already cultivated and 25% grazed • Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil	2
07.2	• Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies • So people with less money lose access to the land and food they used to produce	2
07.3	• It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services • So that pesticides and artificial fertilisers become unnecessary rather than just being banned	2

Question	Marking guidance	Marks
07.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Unsustainable features: reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • Pest control: cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • Nutrients: recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • Water: low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • Soil and carbon: low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • Biodiversity: conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland; agri-environment schemes • Evaluation: growing population (perhaps 12 billion) needs more food; lower trophic level diets; obstacles - cost, need for support, gaining acceptance of new techniques; learning from past land degradation (fertile crescent, US mid-west) • Judgement: no single strategy sufficient; combination of ecological principles and new technology; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	